

Multi-Task Learning for Fire Ignition Maps

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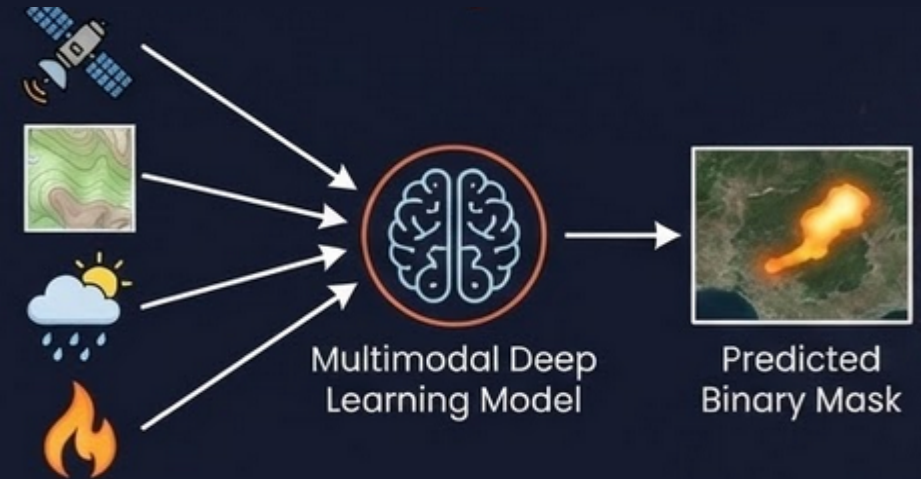
Project Objective

Core Objective

To develop a multimodal deep learning model capable of **Predicting the final burned area extent** right at the moment of ignition, using only pre-fire multimodal data that fused into a multimodal model

Input: Pre-fire environmental data (Satellite, Topography, Weather) + Ignition Point.

Output: Predicted binary mask of the final burned area.



The Paradigm Shift

Current State (Reactive): Remote sensing is primarily used for post-event damage assessment.

Target State (Proactive): To estimate the spread of fire immediately to allocate resources immediately during the "Golden Hour".



Value Proposition



Environmental Impact

Proactive predictions reduce environmental losses by enabling faster response before uncontrolled spread. Fewer hectares burned → lower CO₂ emissions, preserved ecosystems, and reduced long-term regeneration costs.



Response Efficiency

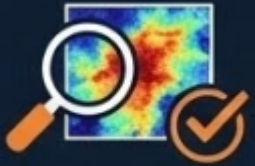
Early burned-area forecasts help Civil Protection allocate firefighting teams, helicopters, and ground units efficiently. Better situational awareness during the first critical hours improves containment and reduces operational risk.



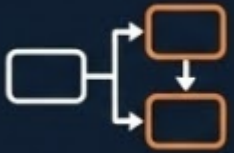
Innovation & Scalability

The multimodal AI system generalizes across regions where Sentinel-2, Landsat, DEM, ERA5, and infrastructure layers are available. The framework can be replicated globally, providing a reusable template for data-driven wildfire risk management.

Research Questions



Can pre-fire multimodal data predict final burned area with sufficient accuracy to support early emergency decisions?



Does auxiliary-task learning improve spatial segmentation performance compared to single-task models?



What is the relative contribution of satellite, meteorological, and topographical modalities?

| Pre-Fire Data:

We will use a catalog of 2152 fires happened in Piemonte region (Italy) in 2012-2024.
For each fire event, the following data are available

Satellite Imagery (Sentinel-2 & Landsat)



Sentinel-2 Imagery: 12-band multispectral, 10-20m res.

- captures immediate **pre-fire condition** (vegetation & fuel availability)



Landsat Imagery: Extra spectral info (SWIR/NIR)

- Sensitive to vegetation moisture and fuel dryness.
- Complementary spectral response for fuel characterization.

Environmental Drivers (DEM & ERA5)



Topograph (DEM): Terrain , slopes and valleys (30m res).

- Fire Behaviour is influenced by **slope and elevation**.



ERA5 weather: Temperature & wind at ignition time.

- Meteorological conditions are key drivers of fire expansion.
- wind affect **direction and speed** of the fire.

Ground-Truth Fire Labels

BURNED AREA MASK & IGNITION DATA



 **Burned Area Mask:** Shape of the area that actually burned, Binary Segmentation (pixel-level supervision)
(1 = burned, 0 = unburned)

why needed? Serve as a **target label** for supervised learning segmentation



 **Ignition Data:** The exact location where the fire started. (Binary marker), derived from Civil Protection point data.

why needed? The model learns how the fire is likely to spread from that specific starting position.

Land cover and Roads

 **Land cover map:** Categorical mask of vegetation types (Forest, Shrub, Grassland, etc.).

why needed? Critical for estimating **fuel type**, Assists the model in differentiating propagation potential based on available fuel.



 **Roads:** Binary road mask (10m resolution).

why needed? Captures both **human-caused** ignition likelihood and fire break potential.



LEARNING PROBLEM DEFINITION: BINARY SEMANTIC SEGMENTATION

INPUT: MULTIMODAL DATA CUBE



OUTPUT: DENSE SPATIAL MASK



Ground Truth/Target

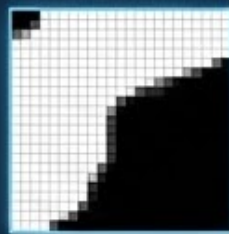


Predicted Segmentation Mask

TASK FORMULATION:
PIXEL-LEVEL
CLASSIFICATION

Each pixel is strictly classified as "Burned" (1) or "Not Burned" (0).

DENSE SPATIAL OUTPUT



Generates a full spatial map, not a single value. Focus is on the entire spatial extent.

LEARNING OBJECTIVE: SPATIAL COHERENCE



Simple Accuracy
(Poor Geometry)



Geometry-Aware
(High Coherence)

Emphasizes capturing the precise shape and boundary, not just pixel counts.

SCIENTIFIC INTENT

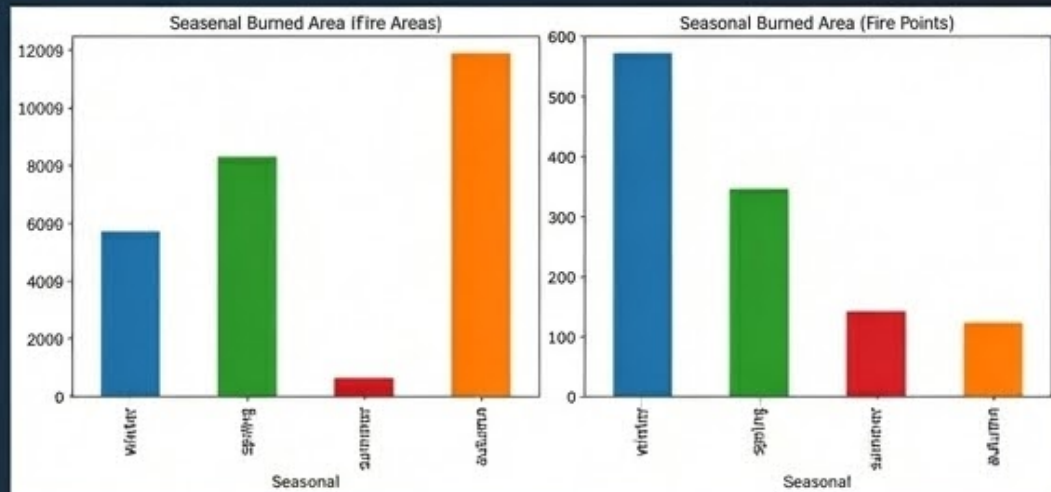
Positioning the work as **dense prediction** (not patch-level) and **geometry-aware segmentation** (not detection).

Dataset Characteristics & Challenges

Scientific Context: Seasonal Patterns & Imbalance



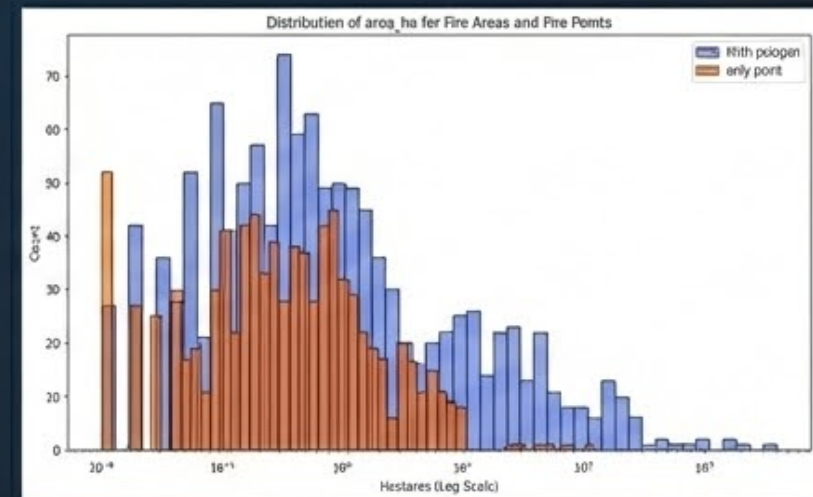
Seasonal Disparities: Frequency vs. Severity



- The number of fires doesn't match how much land gets burned.
- Winter has the most fire starts, but they usually stay small.
- Autumn has fewer fires, but they grow much bigger and burn more land.
- This means we need to predict how **bad** a fire will be, not just if it will start.



Size Distribution & Dataset Imbalance



- Our data is very unbalanced, with a 'long tail' of rare events.
- Most fires are tiny (less than 1 hectare, like a small field).
- The "Polygon" data (blue) shows the rare but huge fires (>100 ha) that cause the most damage.
- This imbalance makes it hard to teach a computer model to predict the big, important fires.

WILDFIRE BURNED-AREA SEGMENTATION ARCHITECTURES

KEY DIFFERENCES (A → B)



FUSION:

Concatenation → Learnable Weights



SUPERVISION:

Single → Multi-Task (Aux)



GOAL:

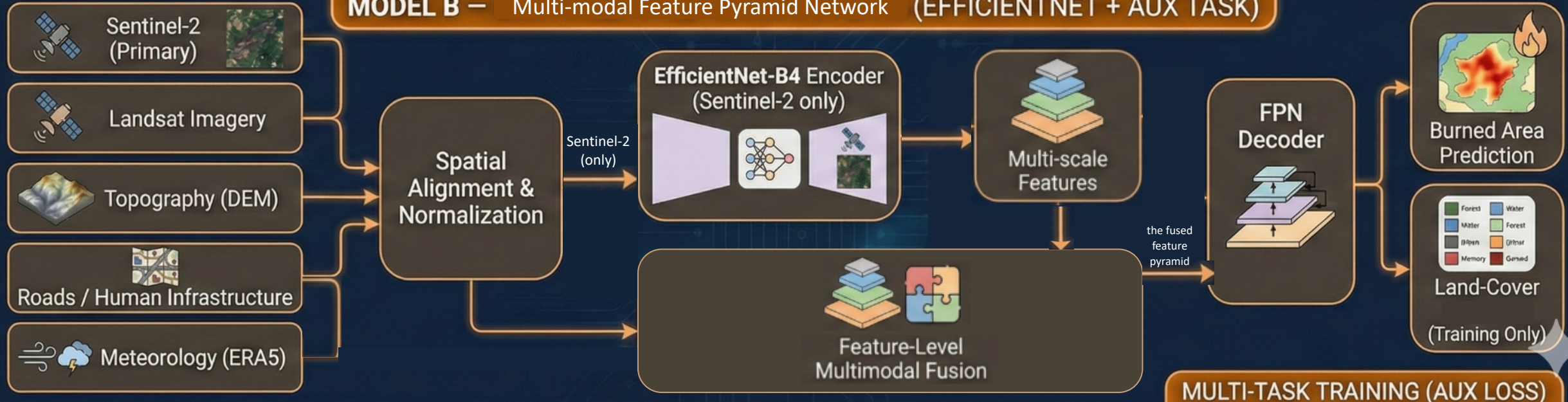
Generalization

MODEL A – BASELINE MULTIMODAL

SINGLE-TASK TRAINING



MODEL B – Multi-modal Feature Pyramid Network (EFFICIENTNET + AUX TASK)



RQ 1: Can **pre-fire** multimodal data predict **final** burned area

we compare a single-task Sentinel-2 baseline against the proposed multimodal multi-task framework. Performance is evaluated on the held-out 2023 test set.



EXPERIMENTAL SETUP

	Parameter	Configuration
	Model	Baseline vs Fusion
	Encoder	ResNet34 vs EFiNet-B4
	Input	Sentinel vs Multimodal
	Epochs	40 vs 120

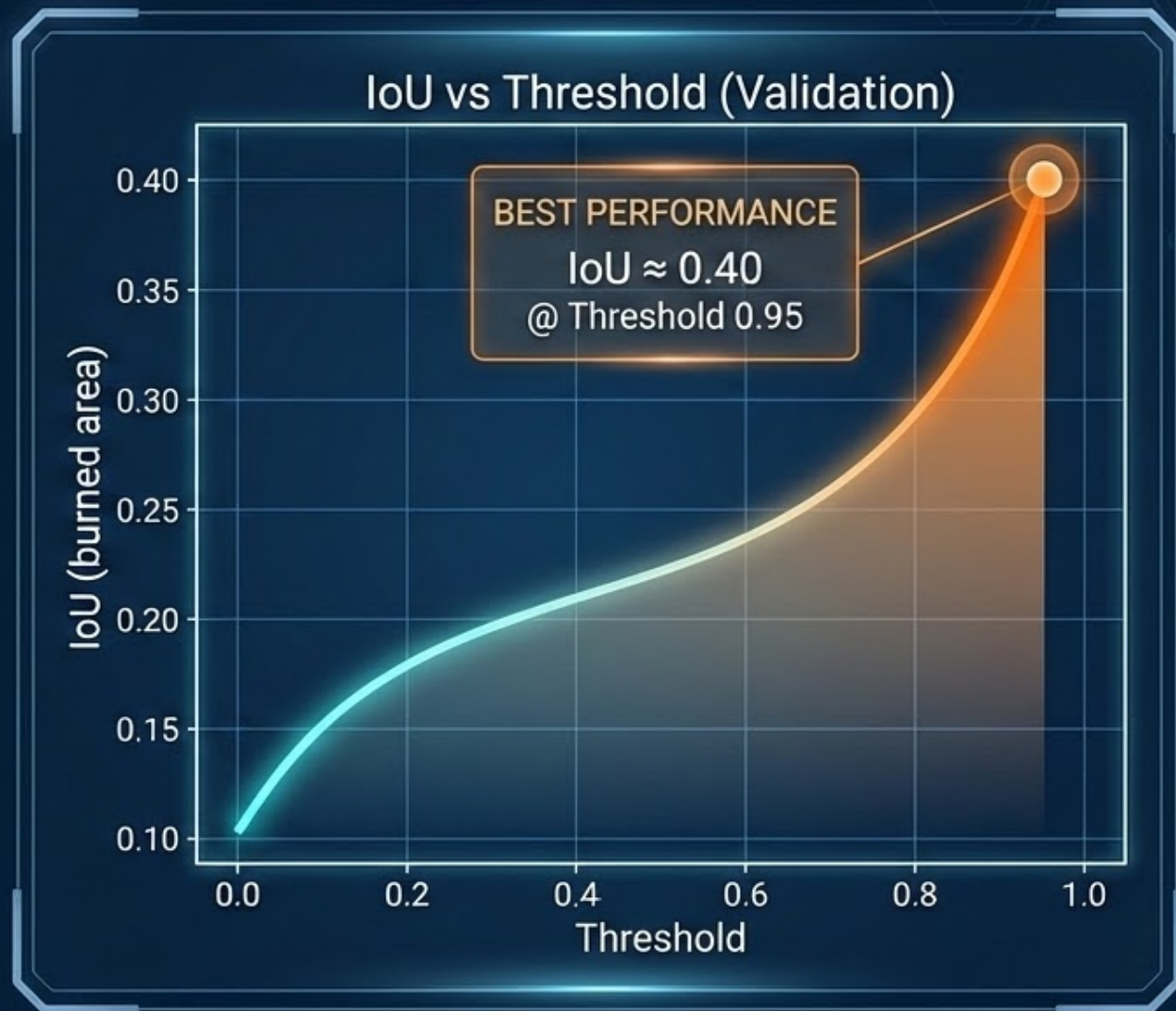
Metric	Baseline	Multimodal
IoU $\tau=0.95$	0.1515	0.4026
F1 $\tau=0.95$	0.2632	0.5741
Precision $\tau=0.95$	0.3468	0.5761
Recall $\tau=0.95$	0.2120	0.5722
IoU $\tau=0.50$	0.1515	0.2341
F1 $\tau=0.50$	0.2632	0.3793

conclusion: Multimodal fusion improves ignition-time burned-area prediction. 70% gain over the baseline.

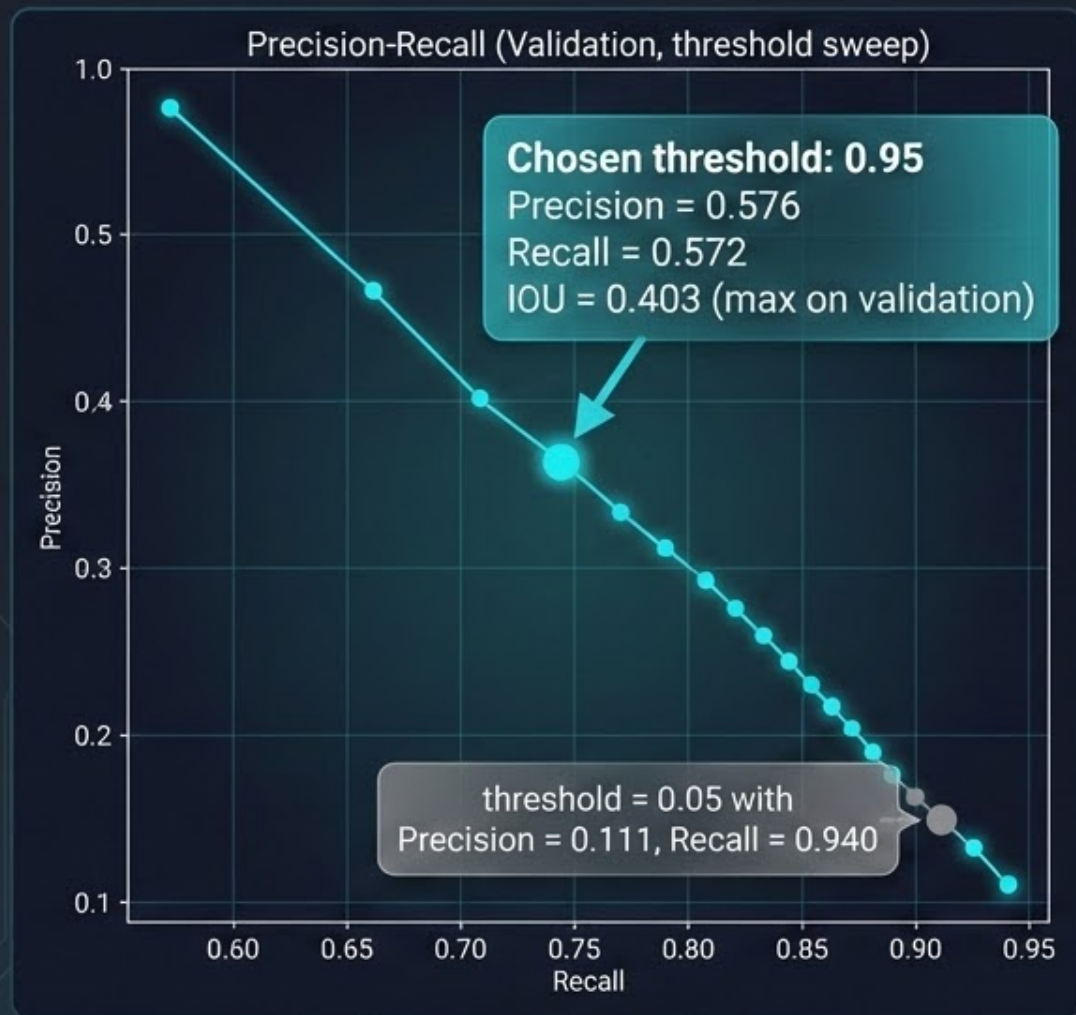
EFFECT OF DECISION THRESHOLD ON IOU

- Default threshold (0.5) significantly underestimates performance.
- IoU increases monotonically with the threshold.
- **Best Performance:** Validation IoU ≈ 0.40 at threshold 0.95.

Threshold calibration improves IoU by **$\sim 70\%$** without retraining.



PRECISION-RECALL TRADE-OFF & THRESHOLD JUSTIFICATION



What the curve means

- 1) Lower threshold** → more 'burned'
↑↓ pixels → recall ↑, false positives ↑ → precision ↓
- 2) Higher threshold** → conservative mask
↑↓ → precision ↑, recall ↓
- 3) Decision:** choose **t = 0.95** because it
✓ maximizes IoU on validation and reduces false alarms

*Validation threshold sweep; pixel-level metrics

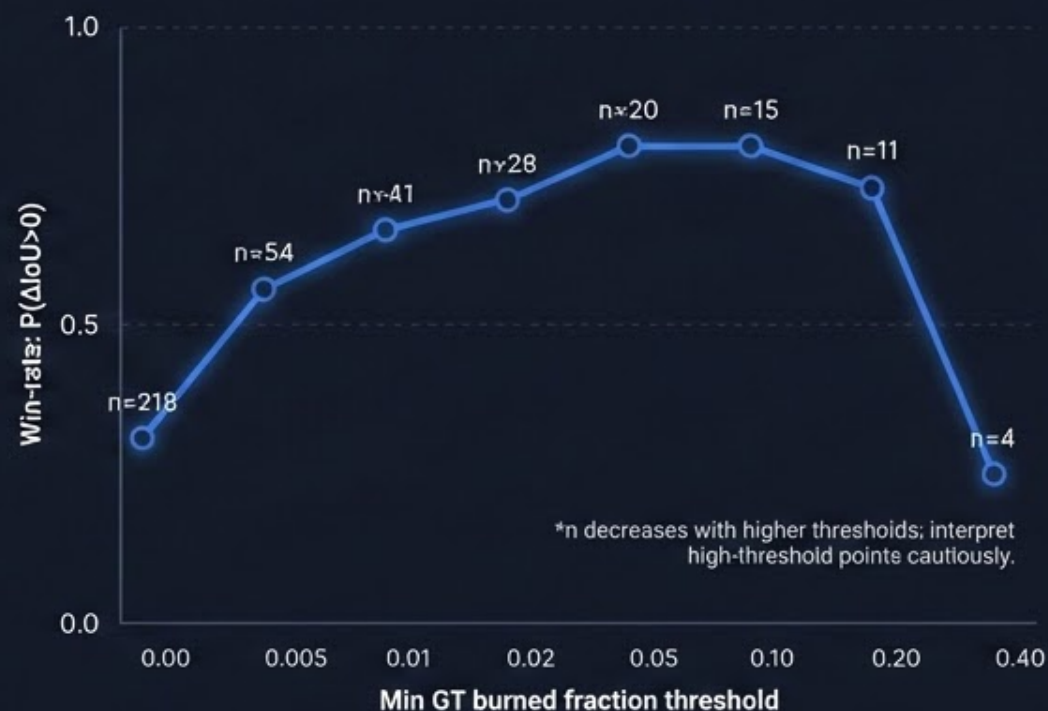
Effect of Adding Auxiliary Land-Segmentation Task in the Multimodal Model

Analysis of ΔIoU by Burned-Fraction Bins and Cumulative Win-Rate Conditioned on Minimum Burn Size (thr=0.95)

Mean ΔIoU by GT Burned-Fraction Bins



Cumulative Win-Rate: NEW (Auxiliary Task) vs. OLD (Baseline)

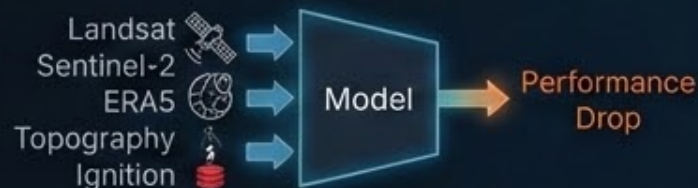


RQ3 – Which modalities contribute most?

🔧 METHOD

We estimate modality contribution using inference-time ablation: run the same trained multimodal model, but **zero-out one modality at a time**.

The performance drop (ΔIoU , ΔF1) indicates contribution.



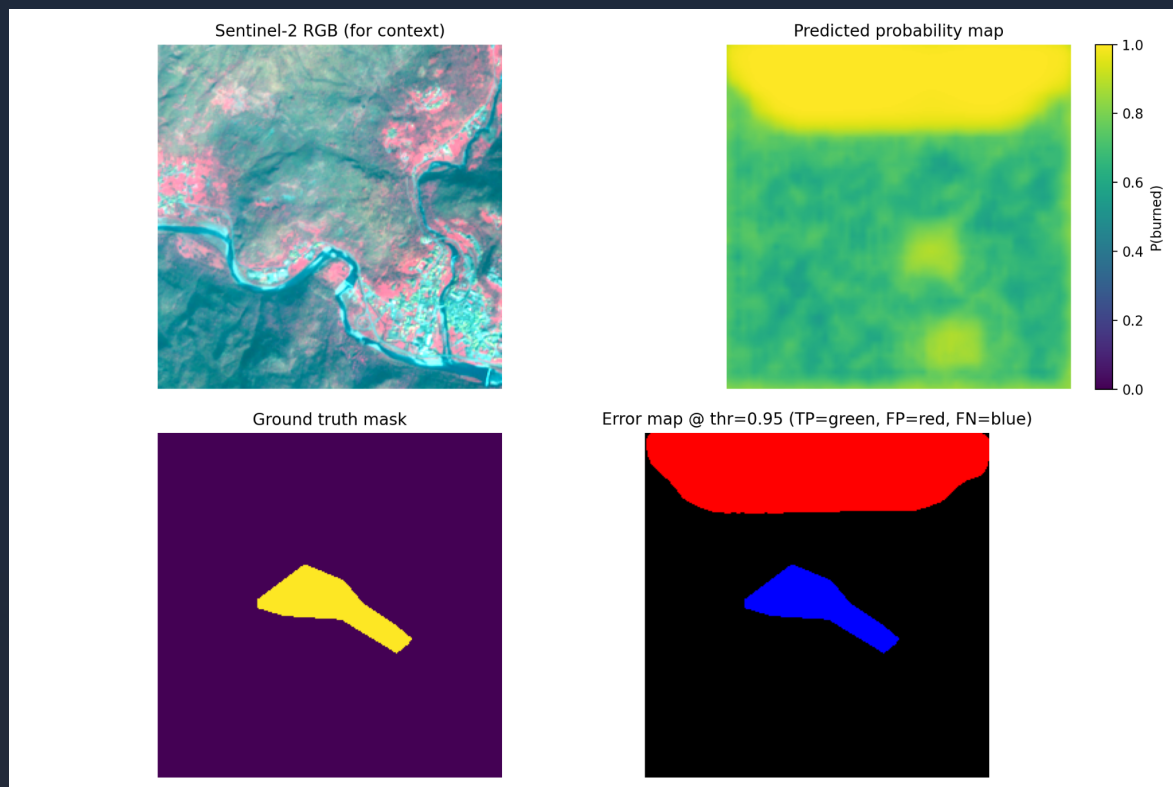
Conclusion: The modality causing the largest drop in IoU/F1 is the most influential.

Sentinel (Ref) shows the highest contribution, followed by Landsat.

Modality Importance: Performance Drop (Higher is Better)

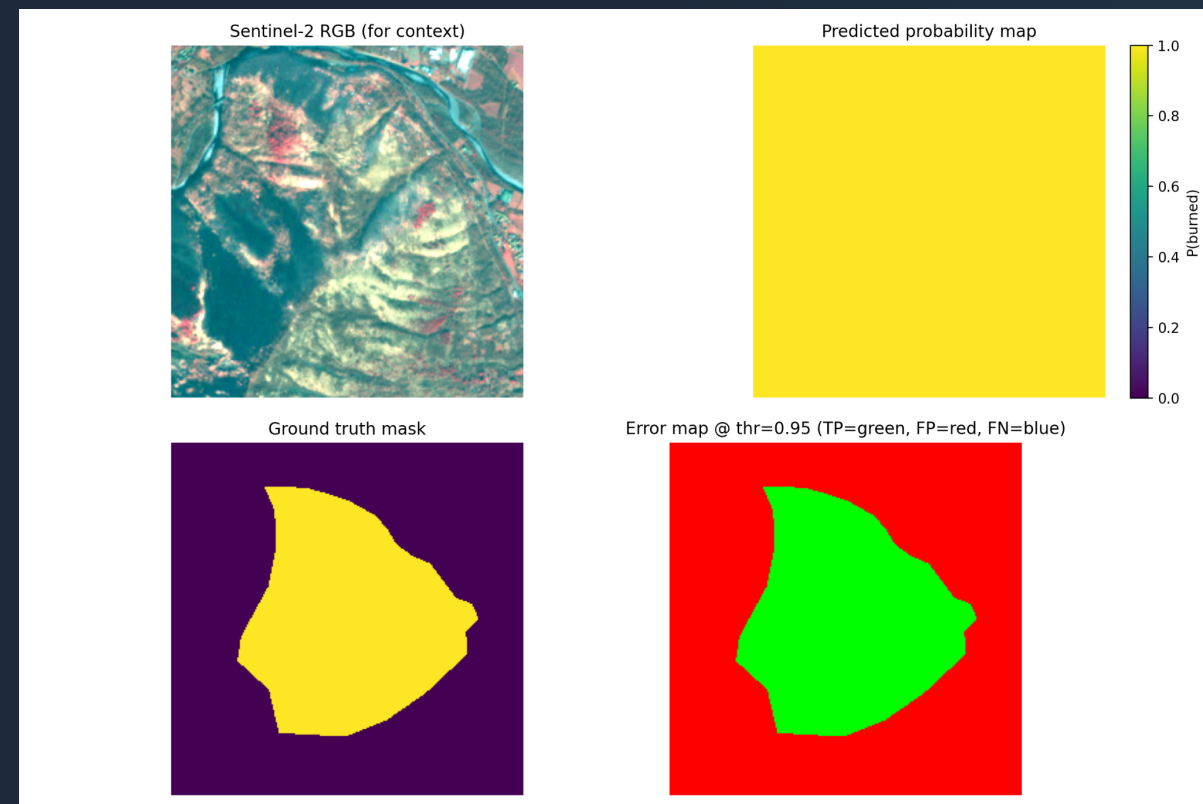


QUALITATIVE EVALUATION



SMALL / DIFFICULT FIRE

Model shows conservative behavior, avoiding noise.



LARGE / WELL-DETECTED FIRE

Large fires are detected with high spatial coherence.

Qualitative Results: Auxiliary task Model Captures Burn Scars More Accurately

Top improvement examples at thr = 0.95 (ΔIoU shown per case)



Case A: idx=125 | $\Delta\text{IoU}=+0.650$ | **Key change:** Auxiliary task recovers missing interior burned area while keeping boundaries tight.



Case B: idx=10 | $\Delta\text{IoU}=+0.644$ | **Key change:** Auxiliary task expands to match GT extent and reduces fragmented false gaps.

What improved (overall)?

- ✓ Higher overlap with GT ($\text{IoU}\uparrow$) by reducing missed burned pixels ($\text{FN}\downarrow$).
- ✓ Better boundary alignment: fewer under-segmented regions than baseline.
- ✓ At thr=0.95, predictions are cleaner and more spatially coherent.

CONCLUSION



Signal Utility

Pre-fire multimodal data provides useful signals for predicting final burned areas.



Encoder Architecture

Encoder design plays a key role: a unified high-capacity encoder achieved better IoU than alternative encoder strategies.



Data Integration

Integrating multiple data sources improves segmentation performance compared to a baseline approach.



Design Balance

Careful architectural choices can improve accuracy while keeping the model pipeline simple and reproducible.



“Overall, this project shows that **effective multimodal integration** combined with a **well-chosen encoder strategy** can significantly enhance burned-area prediction performance.”

THANKS FOR YOUR ATTENTION!



Mission Accomplished!
...or is it?

ANY QUESTIONS?

Disclaimer: Our models are good, but they can't predict *your* questions.